

Calculators may be used in this examination provided they are not capable of being used to store alphabetical information other than hexadecimal numbers



**UNIVERSITY OF
BIRMINGHAM**

School of Computer Science

Advanced Cryptography

Mock Examinations 2025

Time allowed: 2 hours

[Answer all questions]

Note

This paper is marked out of 100.

Question 1

- (a) A colleague is confronted with the concept of secret sharing. Reading up on secret sharing, they discovered a link with one-time encryption, and are now confused regarding several statements (see below).

Comment on the validity of each statement (correct, incorrect/misleading) and explain (briefly) why they are either correct or incorrect/misleading.

- A.** A modern block cipher offers better security guarantees because it has longer keys than the one-time pad.
- B.** Shamir $(2, 2)$ secret sharing of a message m when implemented in $\mathbb{Z}_2$ is the same as Boolean $(2, 2)$ secret sharing in $\mathbb{Z}_2$.
- C.** In the Real vs Random security game, the adversary can ask for the internal randomness of a scheme.
- D.** Perfect indistinguishability of shares implies that an adversary with infinite computing power cannot learn any additional information about the secret after seeing at most t (threshold) shares.

[12 marks]

- (b) Your friend is impressed by your understanding of cryptography. They have come across the concept of Real-Random security for one time encryption, and want your help with it in the context of a clearly insecure encryption scheme that is based on randomly rearranging the letters that represent a message $\mathbf{m}$.

For efficiency reasons, the scheme requires any message to be of even length. Then a one time permutation is sampled from the space of all permutations with length $|\mathbf{m}|/2$, and the ciphertext is derived via the repeated application of the permutation $\mathbf{c}_{k_i} = \mathbf{m}_i$ to both halves of the message. The encryption scheme is defined in the listing below.

For instance, assume the plaintext $abcde fghij$ is meant to be encrypted. Then a valid key would be $\mathbf{k} = \{2, 3, 1, 4, 5\}$, i.e. $\mathbf{k}_1 = 2, \mathbf{k}_2 = 3$, etc., and thus the corresponding ciphertext would be $cabde hfgij$.

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 $\mathcal{K}$  = All Permutations of  $\{1, 2, \dots, |\mathbf{m}|/2\}$ 
def KeyGen( $\mathcal{K}$ ) :
     $\mathbf{k} \xleftarrow{s} \mathcal{K}$ 
    return  $\mathbf{k}$ 
def Enc( $\mathbf{k}, \mathbf{m} = m1 || m2$ ) :
    for  $i = 1$  to  $l/2$  :
         $c1_{\mathbf{k}_i} = m1_i$ 
    for  $i = 1$  to  $l/2$  :
         $c2_{\mathbf{k}_i} = m2_i$ 
    return  $\mathbf{c} = \{c1 || c2\}$ 
def Dec( $\mathbf{k}, \mathbf{c} = c1 || c2$ ) :
    for  $i = 1$  to  $l/2$  :
         $m1_i = c1_{\mathbf{k}_i}$ 
    for  $i = 1$  to  $l/2$  :
         $m2_i = c2_{\mathbf{k}_i}$ 
    return  $\mathbf{m} = \{m1 || m2\}$ 

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Listing 1: Masking based encryption

- Give the Real and Random world for this scheme (as defined in the lectures/hand-out).
- Show via an example how the Real and Random Worlds can be distinguished by an adversary.
- Explain, in your own words, what intuition/security guarantee the Real vs. Random game captures.
- Explain, in your own words, in the context of one-time encryption, what an active and what a passive adversary is, and which adversary the Real vs. Random world assumes.

[12 marks]

- (c) Your friend is now tasked to set up of a secret sharing scheme for their company. The company has s subsidiaries, each with a nominal head, a board with b members, and a single personnel representative. There are more board members than there are subsidiaries, i.e. $b > s$. For major personnel decisions, a majority in a voting like process must be achieved. The voting like process is implemented by reconstructing a secret string, via recombination of sufficiently many shares from a (t, n) secret sharing scheme. The share distribution must satisfy the following properties
- Each head of a subsidiary must at least have one share, and all heads receive an equal number of shares.

- Each board member must have exactly one share.
- The heads of the subsidiaries jointly must have more shares than all board members combined.
- The personnel representative must be able to veto any decision, and therefore must have as many shares as the board members and heads combined.

They wish to share out a secret code that consists of 4 numbers (ranging from $0, \dots, 9$), for instance a code could be 0123.

- A.** How many shares must each of the subsidiaries heads' receive, and how many shares must the personnel representative receive, given the stated constraints? How many shares are given out in total? **[5 marks]**
- B.** Explain how to set up a Shamir secret sharing scheme. Specifically, state what is suitable threshold t given the setting described above, how many shares n must be given out, what is the degree d of the polynomial, what is the size of the modulus? How are the shares generated? **[5 marks]**

Question 2

You have been head-hunted by a large car manufacturer. They now wish to deploy strong cryptography, and at the same time, they wish to secure their wireless key fob against side channel attacks: their key fob stores/uses one of a few static keys. As they are a “forward looking” company, they decided that it is time to (at least) upgrade to using the DES algorithm. They need your expertise to make an informed choice about the implementation of countermeasures.

DES was designed specifically for hardware implementations, and can run in very resource constraint environments. It has a small state (64 bits). At the beginning of a DES round, half of the state is exclusive-ored with a round key (akin to what happens in the AES `AddRoundKey` function). Then the state is expanded and 8 different substitutions $S_i()$ are applied to six bits of the state. The substitutions are “compressing” and give a four bit output. Then the state bits are permuted again. The whole process is depicted in Fig. 1.

- (a) Explain the five steps for a differential input attack. **[15 marks]**
- (b) Reason what side channels are a particular threat to a key fob. **[6 marks]**
- (c) DES has eight different substitution boxes, and, it does bit-wise permutations. What hiding strategies would you recommend ? **[4 marks]**
- (d) The company also approached you with several questions, where they wish clarification:
 - A.** They heard that you need at least a (3,3) Boolean secret sharing scheme to achieve security against differential input attacks. Is this true? What security guarantee(s) does a (3,3) Boolean secret sharing scheme give? **[4 marks]**
 - B.** They are aware of the 3 – *DES* construction (i.e. doing DES three times with either two or three keys), and wonder if 3 – *DES* would increase the number of needed traces for a differential input attack by a factor of 3? **[3 marks]**
 - C.** They also consider strengthening DES by running it with more rounds. Would this make the algorithm harder to attack using side channels? **[3 marks]**

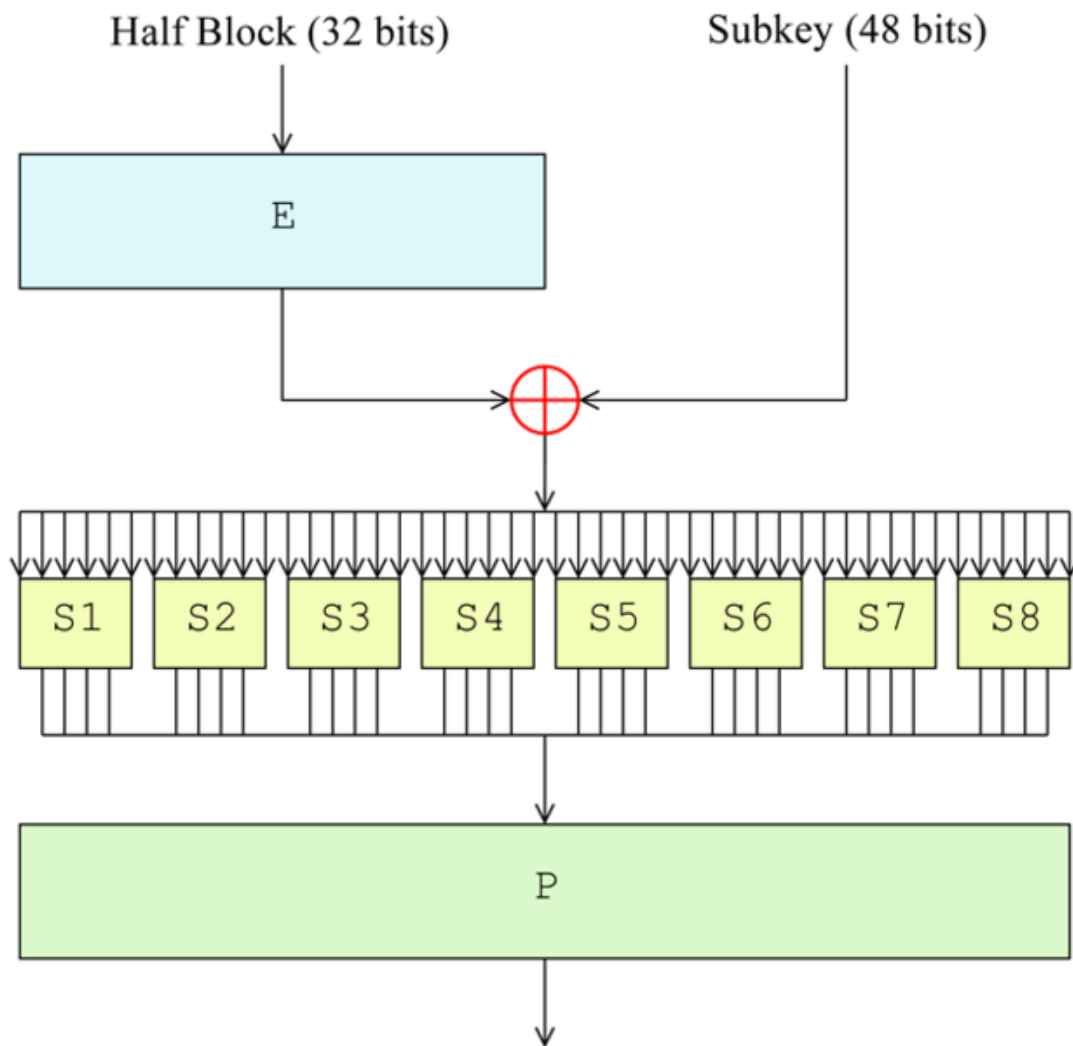


Figure 1: DES round

Question 3

This question is about multi-party computation (MPC).

- (a) What is MPC used for in practice? What security guarantees are expected (consider passive and active adversaries)? **[10 marks]**
- (b) An MPC solution for the private computation over the input values of n clients is being designed using a $(2, 3)$ Shamir secret sharing scheme. The function that is being computed is a weighted sum over the clients inputs m_i : $s = \sum_{i=1}^n i \cdot m_i$. There seems to be a problem with the implementation of the MPC addition in one of the parties. Here is a snap shot of an exemplary computation for $n = 4$ clients. You can see the shares that each party has, and the weighted sum that they have computed. One party's weighted sum is wrong (determine this party). Correct their mistake to compute the final weighted sum. You can use the public recombination vectors. These are for parties $(P_1, P_2) = (2, 22)$, $(P_1, P_3) = (13, 11)$, $(P_2, P_3) = (3, 21)$. All computations happen over $\mathbb{Z}_{23}$: is this a well chosen modulus? **[8 marks]**

Value	P_1	P_2	P_3
m_1	3	5	7
m_2	7	12	17
m_3	4	5	6
m_4	7	10	13
$\sum_i i \cdot m_i$	11	16	19

- (c) Comment on the validity (correct vs incorrect/misleading) of the following statements about the BGW protocol for secure computation. Briefly explain your reasoning.
- A.** The BGW protocol cannot be used when inputs are floating point numbers. **[3 marks]**
- B.** If a multiplication with a constant needs to be carried out, then all parties must communicate to reduce the degree of the resulting polynomial. **[3 marks]**
- E.** The BGW protocol is computationally secure. **[3 marks]**

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.